

DIRECTIONS for questions 1 to 4: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

The definition of a supergene is rather technical, and scientists still argue about its finer points. But at its simplest level a supergene is a group of genes that are inherited together as a unit, often with a lot of other noncoding DNA. You can continue to produce two distinct traits with multiple genes and not worry about them becoming jumbled up. That jumbling often occurs during the production of egg cells and sperm. In that process, the maternal and paternal copies of chromosomes line up and randomly swap segments of DNA in a ballet called recombination. Recombination hedges nature's bets about the value of different permutations of genes; it boosts genetic diversity and helps weed out harmful mutations.

The superpower of supergenes is that they block this. Typically, supergenes contain DNA deletions, insertions or inversions (sequences that were cut out and spliced in backward). As a result, those parts of the chromosomal DNA don't align with a partner and are far less likely to recombine.

In the 1970s, researchers showed that this same mechanism – with misalignments in chromosomes blocking recombination in segments of chromosomes that then continue to lose genes – led to the evolution of Y sex chromosomes from X chromosomes in mammals. Both supergenes and sex chromosomes exist because there's sometimes a benefit to having some sets of genes inherited together. In those cases, it would be ideal to not have any recombination but to have the things that go well together stuck together for good.

To understand why that might be advantageous, think about doing laundry. Say you have a basket of white towels and a basket of red towels. Recombination does the equivalent of tossing both loads into the same drum, flipping on the hot water, and pressing start. What results is a bunch of pink towels...

Many researchers have been probing how supergenes arise and what the consequences for species might be as their supergenes continue to evolve. In one recent effort, Katie Lotterhos, an evolutionary marine biologist at Northeastern University, built a computer model to study the first tentative steps taken on the path from inversion to supergene. Her model showed that the larger the initial DNA flip-flop, the more likely a supergene was to evolve. The reason was simple: A larger inverted fragment of DNA was more likely to capture multiple genes and lock them together as a single entity. Any beneficial mutations arising within the inversion could then promote its spread as a supergene.

But the more important insight from Lotterhos' model was that inversions themselves do not necessarily provide an evolutionary advantage. If a suite of genes is already well adapted to its surroundings, locking it into an inversion will not suddenly allow it to take off as a supergene. That fact may help to explain why complex vital traits aren't routinely secured as supergenes: Ordinary selection pressures are often sufficient to preserve the traits. The question of whether an adaptation precedes an inversion or vice versa, Lotterhos realized, might never be answerable. "What comes first, the inversion or the adaptation?" she said. "It's probably a little bit of both."

Q1. The author uses the laundry analogy to elucidate the point that

- ☐ a) the results of recombination can sometimes be disastrous.
- ☐ b) recombination is counterproductive as it lowers heterogeneity.
- ☒ c) sometimes it pays to keep some combinations of genes apart from each other. **✗ Your answer is incorrect**
- ☐ d) specific sets of genes that offer specific benefits together shouldn't be reshuffled.

Q2. Which of the following is true about a supergene?

- ☐ a) A supergene is formed by pure accident.
- ☐ b) A beneficial mutation inside an inversion is propagated as a supergene.
- ☐ c) **If the suite of genes does not offer any evolutionary benefits, the supergene is stifled.**
- ☐ d) A supergene is propagated whenever it contains a big chunk of genes locked in an inversion.

Q3. The author believes that complex vital traits aren't routinely secured as supergenes because

- ☐ a) inversions of a suite of genes well-adapted to its surroundings is impossible.
- ☐ b) selection pressures prevent recombination of well-adapted genes.
- ☒ c) **ordinary selection preserves only a suite of genes locked in an inversion.** ✖ **Your answer is incorrect**
- ☐ d) the beneficial mutations can be preserved even with ordinary selection.

Q4. All of the following are implied by the arguments presented in the last two paras of the passage EXCEPT:

- ☐ a) Inversions do not occur to specifically provide an evolutionary advantage.
- ☐ b) An inversion with a beneficial suite of genes may take off as a supergene.
- ☐ c) **Genes locked together in an inversion may develop beneficial mutations.**
- ☐ d) Genes once locked together in an inversion always develop beneficial mutations.

DIRECTIONS *for questions 5 to 8:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

As early as 1758, Samuel Richardson had wondered if the novel were not just a fad, whose time had already run out. By the 20th century, the picture looked far bleaker. Theodor Adorno felt that there could be no poetry after Auschwitz. In 1959, Brion Gysin complained that fiction was lagging 50 years behind painting. In the early 60s, Alain Robbe-Grillet attacked the mummification of the novel in its 19th-century incarnation. In 1967, John Barth published "The Literature of Exhaustion" in which he spoke of "the used-upness of certain forms or exhaustion of certain possibilities". The same year, Gore Vidal diagnosed that the novel was already in its death throes: "we shall go on for quite a long time talking of books and writing books, pretending all the while not to notice that the church is empty and the parishioners have gone elsewhere to attend other gods". The death of literature, and the world as we know it, became a fashionable topic among US academics in the early 90s. Their argument was usually that English departments had been hijacked by cultural studies, Continental theory or political correctness gone mad.

Since then, two things have happened. The novel – which was meant to fuse poetry and philosophy, to subsume all other genres and even the entire universe – has been reduced to "literary fiction": a genre that approaches writing as if the 20th century had never happened. At the same time, the digital age has taken information overload to a whole new level. As a result, David Shields believes that the novel is no longer equipped to reflect the vitality and complexity of modern life (Reality Hunger, 2010). Kenneth Goldsmith – the poet to whom we owe the wonder that is UbuWeb – urges us to stop writing altogether in order to focus on recombining the texts we've accumulated over the centuries (Uncreative Writing, 2011). We may all be "remixologists" now, but what if (as Lewis Carroll wondered) word combinations were limited, and we had used them all up?

"Even originality itself no longer has the ability to surprise us," writes Lars Iyer in a remarkable essay recently published by The White Review. According to the author of *Spurious*, we live in "an unprecedented age of words", but one in which Important Novelists have given way to "a legion of keystroke labourers". Literature only survives as literary-fiction kitsch: a "parody of past forms"; a "pantomime of itself". In "The Literature of Exhaustion", Barth had envisaged how the "felt ultimacies of our time" could become the material of future works. Iyer cranks this up a notch. We are no longer writing literature's conclusion but its "epilogue": ours is a "literature which comes after literature". Where Bloom's Romantic poets felt "belated" vis-à-vis their predecessors, Iyer feels that we have come too late for literature, full stop. Literature today is thus no longer "the Thing itself, but about the vanished Thing". The writer's task is "to conjure the ghost" of a tradition that has given it up. By this token, the novels of Tom McCarthy, Lee Rourke and Iyer himself are not so much evidence of a nouveau roman revival as instances of a new type of hauntological fiction which explores the lost futures of Modernism.

Q5. The purpose of this passage is chiefly to

- ☐ a) criticize the quality of novels that are being produced in contemporary times.
- ☐ b) bemoan the demise of literature.
- ☐ c) alert the lovers of literature to the quickly disappearing popularity of literature.
- ☐ d) express angst about the lack of effort on the part of modern-day writers.

Q6. The expression “the church is empty and the parishioners have gone elsewhere to attend other gods” is

- ☐ a) a metaphor for the decline in the quality of novels
- ☐ b) an analogy for the decline in the popularity of the novels
- ☐ **c) an example of how literature has lost its following**
- ☐ d) a comparison between the popularities of literature and other artforms.

Q7. Which of the following summarizes the changes that have happened since the early 90s as mentioned in the second para of the passage?

- ☐ a) The novel has subsumed all other genres to become literary fiction during the digital age.
- ☐ b) The novel is not complex enough to represent the modern world despite the overload of information.
- ☐ c) **The trivialization of the novel has overlapped with the oversaturation of digital information.**
- ☐ d) The novel has ignored the lessons of the 20th century to become shallow and simple.

Q8. Critics of the modern literature have used all of the following expressions to highlight its debilitated state EXCEPT:

- ☐ a) A parody of past forms
- ☐ b) A pantomime of itself
- ☐ c) **A literature which comes after literature.**
- ☐ d) Felt ultimacies of our time

DIRECTIONS *for questions 9 to 12:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

The scale of response to environmental issues is a key point of debate. [Some argue that] localization is one of the most important elements to enacting environmental sustainability... However, our environmental problems are far more complex, and global, than the notion of 'localization' suggests. To understand this, we need to explore the evolution of environmental concern from one of local problems to global environmental issues.

There is no clear beginning to environmental concern. Plato, Lucretius and Caesar all noted the problems of soil erosion. We now know that many of today's most pressing environmental issues were first identified in the nineteenth and early twentieth centuries. Carter (2001) has categorized the evolution of environmental issues into three key periods. Pre-1960s a great deal of environmental concern was focused upon small-scale localized problems such as local pollution and wildlife protection. In the 1960s concern grew about the interrelation between human actions and more complex environmental implications.

At the same time there were growing concerns about so-called 'overpopulation'. Paul Ehrlich, Garrett Hardin and Barry Commoner were considered neo-Malthusian because they saw environmental problems as a consequence of population growth, following the arguments set out by Thomas Malthus in the late eighteenth century. The misguided simplicity of neo-Malthusian explanations of environmental problems is shown in studies that have tried to explain land degradation in terms of population dynamics. Not only is the evidence contradictory, for example population growth in northern China has been linked to increased land degradation, whereas in northern Yemen and Kenya's Machakos Hills it has been related to population decline, but also it is clear from these studies that linking land degradation to one explanatory parameter – population size – masks the complexity behind population dynamics and labour-land relationships. By the early 1970s there was recognition not just of the interconnectedness of environmental issues, and of the problems of cumulative damage, but also that these were global problems that needed internationally coordinated responses.

Environmental scientists identify two types of global environmental change. On the one hand, systemic change occurs when there is a direct impact on a physically interconnected, global system, i.e., the atmosphere or the oceans: for example, the effects that increased emissions of greenhouse gases have on global climate. Cumulative change, on the other hand, occurs when many discrete events become significant because their distribution is global or because, added together, their impact is felt across a large proportion of the globe. For example, across the Americas, Africa and Asia-Pacific, the relatively small amounts of tropical forest lost in each area being converted from forest to farmland, or being lost to a mining operation, add up to a global-scale problem affecting humid tropical forests, one of the most biodiverse biomes found on earth: forests of vital significance to the global water and carbon balances. Recognition of this duality is important in formulating policies and strategies. Those actors addressing systemic issues need to be global, with all countries agreeing and adhering to a particular policy or strategy. Those focusing on cumulative issues require both global approaches and local approaches...

Q9. Which of the following has not been mentioned as one of the focus areas of the three key periods of evolution of environmental issues?

- ☐ a) Problems with localised consequences
- ☐ b) The environmental consequences of human actions
- ☒ c) **Coordinated responses needed for interconnected environmental issues**
- ☐ d) Environmental problems caused by overpopulation

Q10. The examples of land degradation in northern China, Yemen, and Kenya were mentioned in the passage to demonstrate that

- ☐ a) neo-Malthusian explanations of environmental problems oversimplify the evidence.
- ☐ b) there is no causal relationship between population growth and land degradation.
- ☐ **c) land degradation cannot be linked to population size because of the complexity of population dynamics.**
- ☐ d) studies have failed to convincingly explain land degradation in terms of population size.

Q11. Which of the following best summarizes the 'duality' mentioned in the last paragraph of the passage?

- ☐ a) Systemic changes at the level of individual systems combine to provide cumulative changes at a global level.
- ☐ b) Systemic change takes place at a global scale, whereas cumulative changes take place at a local scale.
- ☐ c) **Systemic change happens when an interconnected system is global, whereas cumulative change happens when the distribution of events is global.**
- ☐ d) Systemic change occurs when systems are interconnected, whereas cumulative change happens when systems are discrete.

Q12. The small amounts of tropical forest lost in the Americas, Africa and Asia-Pacific provide a good example for which of the following?

- ☐ a) Discrete events in sufficient numbers can create a global impact.
- ☐ b) Forests are of vital significance to the global water and carbon balances.
- ☐ c) **Humid tropical forests are amongst the most biodiverse biomes found on earth.**
- ☐ d) Those addressing cumulative issues require both global and local approaches.

DIRECTIONS for questions 13 to 16: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

There's been a long-standing debate as to whether neural networks learn in the same way that humans do. Now, a study published last month suggests that natural and artificial neural networks (ANN) learn in similar ways, at least when it comes to language. The researchers – led by Gašper Beguš, a computational linguist at the University of California, Berkeley – compared the brain waves of humans listening to a simple sound to the signal produced by a neural network analyzing the same sound. The results were uncannily alike...The results not only help demystify how ANNs learn, but also suggest that human brains may not come already equipped with hardware and software specially designed for language.

To establish a baseline for the human side of the comparison, the researchers played a single syllable – “bah” – repeatedly in two eight-minute blocks for 14 English speakers and 15 Spanish speakers. While it played, the researchers recorded fluctuations in the average electrical activity of neurons in each listener's brainstem – the part of the brain where sounds are first processed.

In addition, the researchers fed the same “bah” sounds to two different sets of neural networks – one trained on English sounds, the other on Spanish. The researchers then recorded the processing activity of the neural network, focusing on the artificial neurons in the layer of the network where sounds are first analyzed (to mirror the brainstem readings). It was these signals that closely matched the human brain waves.

The researchers chose a kind of neural network architecture known as a generative adversarial network (GAN). A GAN is composed of two neural networks – a discriminator and a generator. The generator creates a sample, which could be an image or a sound. The discriminator determines how close it is to a training sample and offers feedback, resulting in another try from the generator, and so on until the GAN can deliver the desired output.

In this study, the discriminator was initially trained on a collection of either English or Spanish sounds. Then the generator had to find a way of producing them. As a result of this training, the discriminator also got better at distinguishing between real and generated ones. It was at this point, after the discriminator was fully trained, that the researchers played it the “bah” sounds. The team measured the fluctuations in the average activity levels of the discriminator's artificial neurons, which produced the signal so similar to the human brain waves.

This likeness between human and machine activity levels suggested that the two systems are engaging in similar activities. Just as research has shown that feedback from caregivers shapes infant productions of sounds, feedback from the discriminator network shapes the sound productions of the generator network.

The experiment also revealed another interesting parallel between humans and machines. The brain waves showed that the English- and Spanish-speaking participants heard the “bah” sound differently... The brainstem of English speakers responds to the “bah” sound slightly earlier than the brainstem of Spanish speakers, and the GAN trained in English responded to that same sound slightly earlier than the Spanish-trained model. This provided additional evidence, Beguš said, that humans and artificial networks are “likely processing things in a similar fashion.”

Q13. Which of the following can be inferred from the passage?

- ☐ a) Feedback from discriminator networks is aimed to mimic feedback from caregivers.
- ☐ b) Several iterations of feedback from generator to discriminator are needed to train GAN.
- ☐ c) **It is possible to compare the processing activity of different neural networks.**
- ☐ d) It is possible to train neural networks to identify syntactical differences in languages.

Q14. It is implied by the statement “As a result of this training, the discriminator also got better at distinguishing between real and generated ones...” that

- ☐ a) analysis of real and generated sounds involves entirely different processing activities.
- ☐ b) it is possible to differentiate between natural and artificial sounds.
- ☐ c) **natural and artificial neural networks process natural and artificial sounds differently.**
- ☐ d) artificial neural networks do not process natural and artificial sounds similarly.

Q15. “Another interesting parallel between humans and machines” mentioned in the last para of the passage is best summarized as:

- ☐ a) English- and Spanish-trained models of GAN differ in their response speed to specific sounds the same way English- and Spanish-speaking participants do.
- ☐ b) English- and Spanish-trained models of GAN differ in their response time to specific sounds by the same amount as English- and Spanish-speaking participants do.
- ☐ c) **English- and Spanish-trained models of GAN differ in their responses in much the same way as English and Spanish-speaking participants do.**
- ☐ d) English and Spanish-trained models of GAN take at least as much time processing sounds as English-speaking participants do.

Q16. Which of the following, if true, most strengthens the author's conclusion in the first para that "The results ... also suggest that human brains may not come already equipped with hardware and software specially designed for language"?

- ☐ a) Artificial neural networks process sounds in the same way that they have been trained to process.
- ☐ b) Brain waves are mimicked by artificial neural networks once they are trained with specific language sounds.
- ☐ c) **Artificial and natural neural networks demonstrate similar processing of language.**
- ☐ d) Language processing ability in human brains is determined by the training during infancy.

Q17. DIRECTIONS *for questions 17 and 18:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Somewhere in all this, it was thought, there also resided a mysterious élan vital, the force that brought inanimate objects to life.
2. Although chemistry had come a long way in the century that separated Newton and Boyle from Scheele and Priestley and Henry Cavendish, it still had a long way to go.
3. No one knew where this ethereal essence lay, but two things seemed probable: that you could enliven it with a jolt of electricity and that it existed in some substances but not others, which is why we ended up with two branches of chemistry: organic (for those substances that were thought to have it) and inorganic (for those that did not).
4. Right up to the closing years of the eighteenth-century scientists everywhere searched for, and sometimes believed they had actually found, things that just weren't there.

Q18. DIRECTIONS *for questions 17 and 18:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. To begin with, the causes of war and peace run far deeper than the movements of armies and troops, strategy and tactics, into the reasons why armies form and move at all.
2. Contrary to Freud’s conclusion that mankind has a universal “instinct for hatred and destruction,” wars of a continental scale lasting years do not just happen, by chance, circumstance, or instinct.
3. In fact, the wellspring of every war is that which makes us human: our capacity to think abstractly, to conceive, and to create.
4. War is an exclusively human activity: animals eat each other, and clash over mates, dominance, and territory, but absent the capacity to pursue chosen goals and values with an organized commitment, they do not wage war.

Q19. DIRECTIONS *for question 19:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

Analyzed over the long term, manufacturing has proven to be by far the most productivity-intensive sector. This phenomenon seems to fit well with what has been previously asserted concerning this sector's ability to create technological breakthroughs. Since technology is the fundamental variable determining productivity, it is inevitable that the sector that is most intimately linked to technology is also the one with the most productivity-enhancing characteristics. The evidence seems to indicate that the other sectors are only capable of passively dealing with technology. As technology recipients, these sectors inevitably end up showing inferior productivity performance.

- ☐ a) Manufacturing's dependency on technological breakthroughs makes it far more productivity-intensive than other sectors.
- ☒ b) **The constant push for higher productivities is why manufacturing creates more technological breakthroughs than any other sector.** ✖ **Your answer is incorrect**
- ☐ c) **Manufacturing is closely linked to technology and that explains why only this sector produces technological breakthroughs.**
- ☐ d) Manufacturing's higher focus on productivity compared to other sectors is intimately linked with technological breakthroughs.

Q20. DIRECTIONS *for question 20:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. New situations, however, do arise, requiring novel, unstereotyped responses; hence development, in the higher organisms, requires a capacity for experiment and innovation – the social correlates of variation and mutation.
2. Social evolution is an interplay of custom with origination.
3. Evolution in man during recorded time has been social rather than biological: it has proceeded not by heritable variations in the species, but mostly by economic, political, intellectual, and moral innovation transmitted to individuals and generations by imitation, custom, or education.
4. Custom and tradition within a group correspond to type and heredity in the species, and to instincts in the individual; they are ready adjustments to typical and frequently repeated situations.

Q21. DIRECTIONS *for questions 21 and 22:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

The conservative who resists change is as valuable as the radical who proposes it – perhaps as much more valuable as roots are more vital than grafts. It is good that new ideas should be heard, for the sake of the few that can be used; but it is also good that new ideas should be compelled to go through the mill of objection, opposition, and contumely; this is the trial heat which innovations must survive before being allowed to enter the human race. It is good that the old should resist the young, and that the young should prod the old; out of this tension, as out of the strife of the genders and the classes, comes a creative tensile strength, a stimulated development, a secret and basic unity and movement of the whole.

- ☐ a) The strife between the young and the old is not different from the strife between classes and genders, as it leads to a stimulated development.
- ☒ b) **The tension between advocacy of new ideas and the resistance to them prods the development of society.** ✓ **Your answer is correct**
- ☐ c) **Innovation takes place as a result of the tension between various groups – the young and the old, the genders, and the classes.**
- ☐ d) Only those ideas that survive resistance from the old can bring innovation and development in society.

Q22. DIRECTIONS *for questions 21 and 22:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

Whatever merits realism may have as an explanation for real-world politics and as a guide for formulating foreign policy, it is not a popular school of thought in the West. Realism's central message – that it makes good sense for states to selfishly pursue power – does not have broad appeal. It is difficult to imagine a modern political leader openly asking the public to fight and die to improve the balance of power. No European or American leader did so during either the World War or the Cold War. Most people prefer to think of fights between their own state and rival states as clashes between good and evil. Thus, leaders tend to portray war as a moral crusade or an ideological contest, rather than as a struggle for power. Realism is a hard sell.

- ☐ a) Realism loses out to morality as a reason that drives the foreign policy of leaders because it is not popular.
- ☐ b) **Leaders disguise their reasons to pursue power as a moral crusade or an ideological clash.**
- ☐ c) **That realism is not as popular as moral reasons to pursue power explains how leaders sell war.**
- ☐ d) While realism explains real-world politics, it is ill-advisable for politicians to admit it because of its unpopularity.

Q23. DIRECTIONS *for questions 23 and 24:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: **Because ethnographic writing involves a reshaping of experience in order to address anthropological conversations, all ethnographers need to consider the gap between the text and the lived reality that they try to explain.**

Paragraph: ____ (1) _____. There is a tension between the chaos and diversity of experience and the transcription of that experience in a text, between life and analysis, that every author must deal with. Moreover, because ethnographers set out to make anthropological claims, because they draw on anthropological concepts and discussions, there is always a distance between the knowledge of experience put forward in ethnography and the local ways of knowing and making sense of the world that the ethnography is trying to explain. ____ (2) _____. All ethnography, then, is moulded by the inevitability of dealing with the gaps between life and text, and between local and anthropological perspectives. ____ (3) _____. And all ethnographers need to make decisions as to how to represent these gaps and how to bridge them. ____ (4) _____.

- ☐ a) Option 1
- ☒ b) Option 2 **✗ Your answer is incorrect**
- ☐ c) Option 3
- ☐ d) Option 4

Q24. DIRECTIONS *for questions 23 and 24:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.
Sentence: **But for all the waste they lay to their prey, pangolins are almost entirely harmless to any other living creature.**

Paragraph: _____(1)_____. When this bashful animal is frightened, it curls up into a tight ball, the scales offering its first – and, really, only – line of defense. _____(2)_____. Sadly, it's that defense mechanism that makes them so susceptible to capture by traffickers. As poachers drive pangolins into the open by setting fire to their burrows or by battering them out of trees with sticks and clubs, the stunned animals roll up and become as compact as a medicine ball, making them easy to scoop up and shove in a sack. _____(3)_____. Often, when poachers are interested only in the scales, they drop the curled-up pangolins into pots of water and boil them alive to make the scales easier to pluck. _____(4)_____.

- ☐ a) Option 1
- ☒ b) **Option 2** ✖ **Your answer is incorrect**
- ☐ c) **Option 3**
- ☐ d) Option 4